HOMEWORK 3.6 · UNIT 3 · NC.8.EE.7

Multi-Step Equations (Day 4)

Oct 20 / Oct 21 · Targets: EE.7-3, EE.7-1, EE.7-2, EE.7-5, EE.7-9

Name _____ Date _____ Block _____

Remember: Plan: distribute → clear fractions or decimals → combine like terms → variable on one side → undo + or − → undo × or ÷ → **check**. Decimals: multiply every term by 10 or 100.

Example: $0.4x + 1.2 = 2 \rightarrow \times 10$: $4x + 12 = 20 \rightarrow 4x = 8 \rightarrow x = 2$

SCORE 3.0 Approaching · the basics

Targets: EE.7-3, EE.7-1, EE.7-2

1 Solve. $0.5x + 2.5 = 4$ EE.7-3

2 Solve. $-3(x + 2) = x - 14$ EE.7-3

3 Which equation has the same solution as $\frac{x}{2} + \frac{1}{3} = 2$? EE.7-3

(A) $3x + 2 = 12$
(B) $3x + 1 = 12$
(C) $3x + 2 = 2$
(D) $x + 2 = 12$

4 Is $x = 9$ a solution of $\frac{x}{3} + 1 = 4$? REVIEW 3.5 EE.7-1

Yes No

Why? _____

5 Solve. $2(x + 3) - 4 = x + 8$ REVIEW 3.4 EE.7-2

SCORE 4.0 Proficient · on target

Targets: EE.7-5

6 **Sofia** made a mistake. Circle it. Explain it. Fix it. EE.7-5

SOFIA'S WORK

$0.3x - 0.6 = 1.5$

$\times 10$: $3x - 0.6 = 15$

$3x = 15.6$

$x = 5.2$

7 Solve. $\frac{1}{4}(x + 8) = \frac{1}{2}x - 1$ EE.7-5

Ⓐ $x = 36$ Ⓑ $x = 12$ Ⓒ $x = -12$ Ⓓ $x = 3$

8 Check by substitution. Is $x = -5$ a solution of $2(x + 7) = -x + 3$? EE.7-5

Yes No

Why? _____

SCORE 5.0 Mastery · go deeper

Targets: EE.7-9

9 Gym A costs \$25 plus \$10 a month. Gym B costs \$5 plus \$14 a month. Solve $25 + 10m = 5 + 14m$. What does it mean? EE.7-9

m = _____. This means _____.

10 Explain why $0.5x = 3$ and $\frac{1}{2}x = 3$ have the same solution. EE.7-9

They have the same solution because _____.